

35 Chapter Review



➔ Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

SUMMARY OF KEY CONCEPTS

➔ To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zkz9t.

CONCEPT 35.1

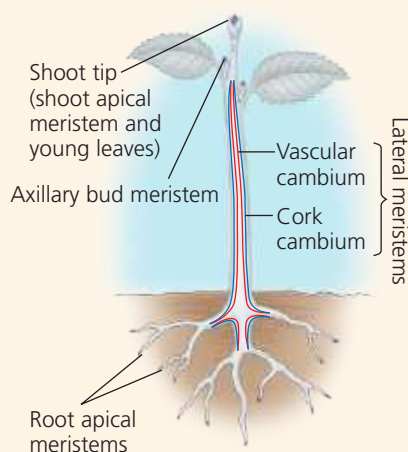
Plants have a hierarchical organization consisting of organs, tissues, and cells (pp. 759–765)

- Vascular plants have shoots consisting of **stems**, **leaves**, and, in angiosperms, **flowers**. **Roots** anchor the plant, absorb and conduct water and minerals, and store food. Leaves are attached to stem **nodes** and are the main **organs** of photosynthesis. The **axillary buds**, in axils of leaves and stems, give rise to branches. Plant organs may be adapted for specialized functions.
- Vascular plants have three **tissue systems**—dermal, vascular, and ground—which are continuous throughout the plant. The **dermal tissue** is a continuous layer of cells that covers the plant exterior. **Vascular tissues (xylem and phloem)** facilitate the long-distance transport of substances. **Ground tissues** function in storage, metabolism, and regeneration.
- Parenchyma cells** are relatively undifferentiated and thin-walled cells that retain the ability to divide; they perform most of the metabolic functions of synthesis and storage. **Collenchyma cells** have unevenly thickened walls; they support young, growing parts of the plant. **Sclerenchyma cells**—sclereids and fibers—have thick, lignified walls that help support mature, nongrowing parts of the plant. **Tracheids** and **vessel elements**, the water-conducting cells of xylem, have thick walls and are dead at functional maturity. **Sieve-tube elements** are living but highly modified cells that are largely devoid of internal organelles; they function in the transport of sugars through the phloem of angiosperms.

? Describe at least three specializations in plant organs and plant cells that are adaptations to life on land.

CONCEPT 35.2

Different meristems generate new cells for primary and secondary growth (pp. 766–767)

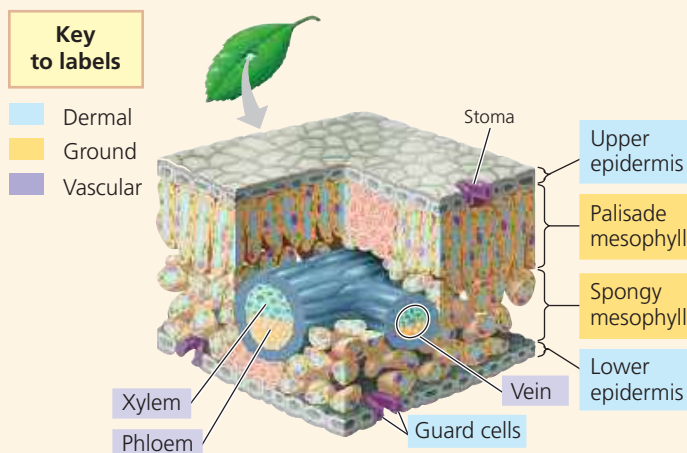


? What is the difference between primary and secondary growth?

CONCEPT 35.3

Primary growth lengthens roots and shoots (pp. 768–771)

- The root **apical meristem** is located near the tip of the root, where it generates cells for the growing root axis and the **root cap**.
- The apical meristem of a shoot is located in the **apical bud**, where it gives rise to alternating **internodes** and leaf-bearing nodes.
- Eudicot stems have vascular bundles in a ring, whereas monocot stems have scattered vascular bundles.
- Mesophyll** cells are adapted for photosynthesis. **Stomata**, epidermal pores formed by pairs of **guard cells**, allow for gaseous exchange and are major avenues for water loss.



? How does branching in roots differ from that in stems?

CONCEPT 35.4

Secondary growth increases the diameter of stems and roots in woody plants (pp. 772–775)

- The **vascular cambium** is a meristematic cylinder that produces secondary xylem and secondary phloem during **secondary growth**. Older layers of secondary xylem (heartwood) become inactive, whereas younger layers (sapwood) still conduct water.

